

Beavergnaw

Tags:

Time Limit: 1 s Memory Limit: 128 MB
Submission: 21 AC: 17 Score: 99.49

Submit

Codes

AI Code Tips

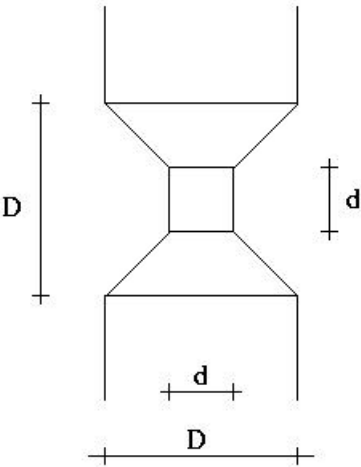
Description

Problem C: Beavergnaw

When chomping a tree the beaver cuts a very specific shape out of the tree trunk. What is left in the tree trunk looks like two frustums of a cone joined by a cylinder with the diameter the same as its height. A very curious beaver tries not to demolish a tree but rather sort out what should be the diameter of the cylinder joining the frustums such that he chomped out certain amount of wood. You are to help him to do the calculations.



We will consider an idealized beaver chomping an idealized tree. Let us assume that the tree trunk is a cylinder of diameter D and that the beaver chomps on a segment of the trunk also of height D . What should be the diameter d of the inner cylinder such that the beaver chomped out V cubic units of wood?



Input contains multiple cases each presented on a separate line. Each line contains two integer numbers D and V separated by whitespace. D is the linear units and V is in cubic units. V will not exceed the maximum volume of wood that the beaver can chomp. A line with $D=0$ and $V=0$ follows the last case.

For each case, one line of output should be produced containing one number rounded to three fractional digits giving the value of d measured in linear units.

Samples

input	Copy
10 250	
20 2500	
25 7000	
50 50000	
0 0	
output	Copy

8.054
14.775
13.115
30.901

Submit

Codes

[HZNUOJ](#) is based on [HUSTOJ](#)

[--- Maintainer List ---](#)

Server Time: 2025-8-26 15:05:32